

Computer Networks

Assignment – 02

IP Addressing, Routing and Congestion Control

Name:	Pranav Sood
UID:	24BDA70188
Section:	24AIT-KRG-2
Subject:	Computer Networks
Subject Code:	24CSH-337
Submitted To:	Mr. Alok
Assignmemnt:	2

Q1. IP Addressing, Classes, Subnetting and Supernetting

IP Address

An IP address is a logical address assigned to a device on a network. It identifies the network and the host so that packets can be delivered correctly.

IPv4 uses a 32-bit address, written as four octets.

192.168.1.10

Binary form:

11000000.10101000.00000001.00001010

An IPv4 address consists of a Network ID and a Host ID.

Why Classes are Used in IPv4

Classful addressing divides IPv4 addresses into different classes according to network size. This allowed organizations to receive networks suitable for the number of hosts they required.

Class A/B/C	Network	Host
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IPv4 Classes

Class	First Octet	Network Bits	Host Bits	Hosts/Network
A	1-126	8	24	$2^{24} - 2 = 16,777,214$
B	128-191	16	16	$2^{16} - 2 = 65,534$
C	192-223	24	8	$2^8 - 2 = 254$
D	224-239	–	–	Multicast
E	240-255	–	–	Experimental

Number of networks:

Class	Number of Networks
A	$2^7 - 2 = 126$
B	$2^{14} = 16,384$
C	$2^{21} = 2,097,152$

Default masks are:

Class A: 255.0.0.0 (/8)

Class B: 255.255.0.0 (/16)

Class C: 255.255.255.0 (/24)

Subnetting

Subnetting divides one large network into smaller networks by borrowing bits from the host portion.

Example:

192.168.1.0/24 → 192.168.1.0/26

Two host bits are borrowed:

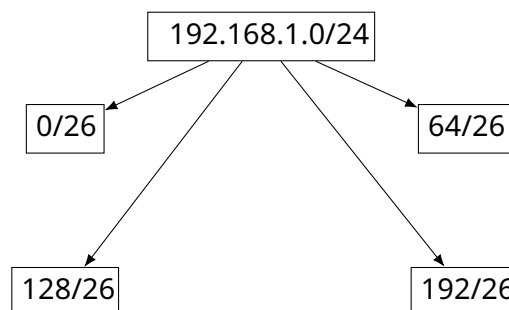
$$2^2 = 4 \text{ subnets}$$

Each subnet contains:

$$2^6 = 64 \text{ addresses}$$

and:

$$64 - 2 = 62 \text{ usable hosts}$$



Supernetting

Supernetting combines multiple smaller networks into one larger network. It reduces routing table entries.

Example:

192.168.0.0/24

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

can be combined as:

192.168.0.0/22

Thus, subnetting divides networks, whereas supernetting combines networks.

Q2. Link State and Distance Vector Routing

Routing determines the path through which packets travel from source to destination. The path is selected using a cost metric such as hops, delay or bandwidth.

Distance Vector Routing

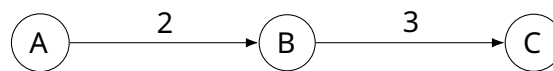
In Distance Vector routing, each router maintains the cost to reach destinations through its neighboring routers.

It uses the Bellman-Ford principle:

$$D_x(y) = \min_v [c(x, v) + D_v(y)]$$

Routers exchange their routing tables with neighboring routers.

Example:



Cost from A to C through B:

$$2 + 3 = 5$$

Therefore:

$$A \rightarrow B \rightarrow C$$

is selected with cost 5.

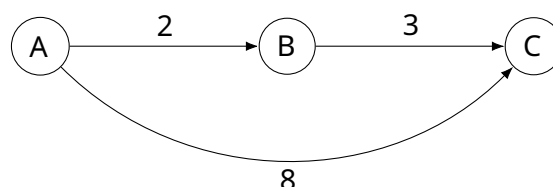
Link State Routing

In Link State routing, each router builds a complete map of the network and calculates the shortest path using Dijkstra's algorithm.

Steps:

1. Discover neighboring routers.
2. Determine link costs.
3. Flood link-state information.
4. Build the network topology.
5. Run Dijkstra's algorithm.

Example:



Two paths exist:

$$A \rightarrow C = 8$$

$$A \rightarrow B \rightarrow C = 2 + 3 = 5$$

Hence Dijkstra selects:

$A \rightarrow B \rightarrow C$

with minimum cost 5.

Difference

Feature	Distance Vector	Link State
Knowledge	Neighbor information	Complete topology
Algorithm	Bellman-Ford	Dijkstra
Updates	To neighbors	Flooded through network
Convergence	Slower	Faster
Example	RIP	OSPF

Q3. Congestion, Traffic Shaping and Fragmentation

Network Congestion

Congestion occurs when the amount of traffic entering a network is greater than the network's capacity to process it.

It may cause:

- Packet loss
- Increased delay
- Retransmissions
- Reduced throughput

For example, if a router can process 1000 packets/s but receives 1500 packets/s, packets start accumulating in its buffer.

Need for Congestion Control

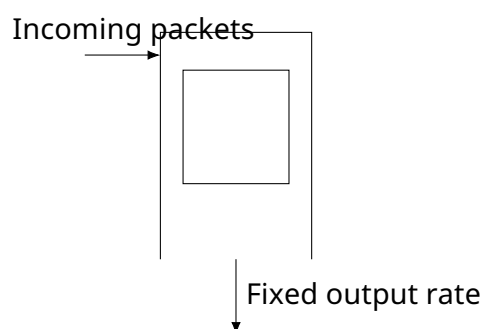
Congestion control regulates traffic so that network resources are not overloaded.

Its objectives are to:

- Reduce packet loss and delay.
- Prevent buffer overflow.
- Maintain good throughput.
- Improve network stability.

Leaky Bucket

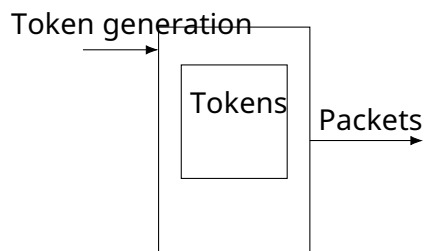
The Leaky Bucket algorithm sends packets at a fixed rate, similar to water leaving a bucket through a small hole.



Example: If the output rate is 2 packets/s, packets leave at approximately 2 packets/s even if they arrive in a burst. If the bucket is full, additional packets are discarded.

Token Bucket

In Token Bucket, tokens are generated at a fixed rate, a packet requires a token before transmission.



Unused tokens can accumulate. Therefore, when a burst arrives, the stored tokens allow multiple packets to be transmitted quickly.

Leaky Bucket vs Token Bucket

	Leaky Bucket	Token Bucket
Output	Fixed rate	Variable rate
Bursts	Not handled well	Allows controlled bursts
Overflow	Packets may be dropped	Extra tokens may be dropped
Main use	Smooth traffic	Flexible traffic control

Fragmentation

Fragmentation is the process of dividing a large IP packet into smaller fragments when the packet is larger than the MTU of the next network.

For example:

Packet = 4000 bytes

MTU = 1500 bytes

Assuming a 20 byte IP header, maximum payload per fragment is:

$$1500 - 20 = 1480 \text{ bytes}$$

The 3980 byte payload can therefore be divided as:

$$1480 + 1480 + 1020 = 3980$$

Thus, three fragments are created.

Fragmentation is required when different networks have different MTU sizes.

Conclusion

IPv4 uses 32-bit addresses to identify networks and hosts. Classful addressing divides the address space into classes, while subnetting divides networks and supernetting combines them.

Distance Vector and Link State routing use different approaches to find least-cost paths. Congestion control prevents excessive traffic from degrading network performance, while Leaky Bucket and Token Bucket regulate traffic. Fragmentation allows large IP packets to cross networks with smaller MTUs.